



## Security Configuration Assessment Report for gn-VirtualBox

Target IP Address: localhost

### CIS Ubuntu Linux 20.04 LTS Benchmark v3.0.0

Level 1 - Workstation  
Friday, August 8 2025 00:24:00  
Assessment Duration: 13 seconds

Report generated by the Center for Internet Security's Configuration Assessment Tool (CIS-CAT Pro Assessor) v4.55.0.

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## Summary

| Description                                                              | Tests     |           |          |          |          |          | Scoring     |             |            |
|--------------------------------------------------------------------------|-----------|-----------|----------|----------|----------|----------|-------------|-------------|------------|
|                                                                          | Pass      | Fail      | Error    | Unkn.    | Man.     | Exc.     | Score       | Max         | Percent    |
| <b>1 Initial Setup</b>                                                   | <b>37</b> | <b>12</b> | <b>0</b> | <b>0</b> | <b>4</b> | <b>0</b> | <b>37.0</b> | <b>49.0</b> | <b>76%</b> |
| 1.1 <a href="#">Filesystem</a>                                           | 23        | 3         | 0        | 0        | 1        | 0        | 23.0        | 26.0        | 88%        |
| 1.1.1 <a href="#">Configure Filesystem Kernel Modules</a>                | 3         | 2         | 0        | 0        | 1        | 0        | 3.0         | 5.0         | 60%        |
| 1.1.2 <a href="#">Configure Filesystem Partitions</a>                    | 20        | 1         | 0        | 0        | 0        | 0        | 20.0        | 21.0        | 95%        |
| 1.1.2.1 <a href="#">Configure /tmp</a>                                   | 3         | 1         | 0        | 0        | 0        | 0        | 3.0         | 4.0         | 75%        |
| 1.1.2.2 <a href="#">Configure /dev/shm</a>                               | 4         | 0         | 0        | 0        | 0        | 0        | 4.0         | 4.0         | 100%       |
| 1.1.2.3 <a href="#">Configure /home</a>                                  | 2         | 0         | 0        | 0        | 0        | 0        | 2.0         | 2.0         | 100%       |
| 1.1.2.4 <a href="#">Configure /var</a>                                   | 2         | 0         | 0        | 0        | 0        | 0        | 2.0         | 2.0         | 100%       |
| 1.1.2.5 <a href="#">Configure /var/tmp</a>                               | 3         | 0         | 0        | 0        | 0        | 0        | 3.0         | 3.0         | 100%       |
| 1.1.2.6 <a href="#">Configure /var/log</a>                               | 3         | 0         | 0        | 0        | 0        | 0        | 3.0         | 3.0         | 100%       |
| 1.1.2.7 <a href="#">Configure /var/log/audit</a>                         | 3         | 0         | 0        | 0        | 0        | 0        | 3.0         | 3.0         | 100%       |
| 1.2 <a href="#">Package Management</a>                                   | 0         | 0         | 0        | 0        | 3        | 0        | 0.0         | 0.0         | 0%         |
| 1.2.1 <a href="#">Configure Package Repositories</a>                     | 0         | 0         | 0        | 0        | 2        | 0        | 0.0         | 0.0         | 0%         |
| 1.2.2 <a href="#">Configure Package Updates</a>                          | 0         | 0         | 0        | 0        | 1        | 0        | 0.0         | 0.0         | 0%         |
| 1.3 <a href="#">Mandatory Access Control</a>                             | 3         | 0         | 0        | 0        | 0        | 0        | 3.0         | 3.0         | 100%       |
| 1.3.1 <a href="#">Configure AppArmor</a>                                 | 3         | 0         | 0        | 0        | 0        | 0        | 3.0         | 3.0         | 100%       |
| 1.4 <a href="#">Configure Bootloader</a>                                 | 0         | 2         | 0        | 0        | 0        | 0        | 0.0         | 2.0         | 0%         |
| 1.5 <a href="#">Configure Additional Process Hardening</a>               | 4         | 1         | 0        | 0        | 0        | 0        | 4.0         | 5.0         | 80%        |
| 1.6 <a href="#">Configure Command Line Warning Banners</a>               | 6         | 0         | 0        | 0        | 0        | 0        | 6.0         | 6.0         | 100%       |
| 1.7 <a href="#">Configure GNOME Display Manager</a>                      | 1         | 6         | 0        | 0        | 0        | 0        | 1.0         | 7.0         | 14%        |
| <b>2 Services</b>                                                        | <b>36</b> | <b>3</b>  | <b>0</b> | <b>0</b> | <b>1</b> | <b>0</b> | <b>36.0</b> | <b>39.0</b> | <b>92%</b> |
| 2.1 <a href="#">Configure Server Services</a>                            | 17        | 0         | 0        | 0        | 1        | 0        | 17.0        | 17.0        | 100%       |
| 2.2 <a href="#">Configure Client Services</a>                            | 4         | 2         | 0        | 0        | 0        | 0        | 4.0         | 6.0         | 67%        |
| 2.3 <a href="#">Configure Time Synchronization</a>                       | 6         | 0         | 0        | 0        | 0        | 0        | 6.0         | 6.0         | 100%       |
| 2.3.1 <a href="#">Ensure time synchronization is in use</a>              | 1         | 0         | 0        | 0        | 0        | 0        | 1.0         | 1.0         | 100%       |
| 2.3.2 <a href="#">Configure systemd-timesyncd</a>                        | 2         | 0         | 0        | 0        | 0        | 0        | 2.0         | 2.0         | 100%       |
| 2.3.3 <a href="#">Configure chrony</a>                                   | 3         | 0         | 0        | 0        | 0        | 0        | 3.0         | 3.0         | 100%       |
| 2.4 <a href="#">Job Schedulers</a>                                       | 9         | 1         | 0        | 0        | 0        | 0        | 9.0         | 10.0        | 90%        |
| 2.4.1 <a href="#">Configure cron</a>                                     | 8         | 1         | 0        | 0        | 0        | 0        | 8.0         | 9.0         | 89%        |
| 2.4.2 <a href="#">Configure at</a>                                       | 1         | 0         | 0        | 0        | 0        | 0        | 1.0         | 1.0         | 100%       |
| <b>3 Network</b>                                                         | <b>10</b> | <b>1</b>  | <b>0</b> | <b>0</b> | <b>1</b> | <b>0</b> | <b>10.0</b> | <b>11.0</b> | <b>91%</b> |
| 3.1 <a href="#">Configure Network Devices</a>                            | 0         | 0         | 0        | 0        | 1        | 0        | 0.0         | 0.0         | 0%         |
| 3.2 <a href="#">Configure Network Kernel Modules</a>                     | 0         | 0         | 0        | 0        | 0        | 0        | 0.0         | 0.0         | 0%         |
| 3.3 <a href="#">Configure Network Kernel Parameters</a>                  | 10        | 1         | 0        | 0        | 0        | 0        | 10.0        | 11.0        | 91%        |
| <b>4 Host Based Firewall</b>                                             | <b>17</b> | <b>8</b>  | <b>0</b> | <b>0</b> | <b>5</b> | <b>0</b> | <b>17.0</b> | <b>25.0</b> | <b>68%</b> |
| 4.1 <a href="#">Configure a single firewall utility</a>                  | 0         | 1         | 0        | 0        | 0        | 0        | 0.0         | 1.0         | 0%         |
| 4.2 <a href="#">Configure UncomplicatedFirewall</a>                      | 3         | 4         | 0        | 0        | 1        | 0        | 3.0         | 7.0         | 43%        |
| 4.3 <a href="#">Configure nftables</a>                                   | 8         | 0         | 0        | 0        | 2        | 0        | 8.0         | 8.0         | 100%       |
| 4.4 <a href="#">Configure iptables</a>                                   | 6         | 3         | 0        | 0        | 2        | 0        | 6.0         | 9.0         | 67%        |
| 4.4.1 <a href="#">Configure iptables software</a>                        | 2         | 1         | 0        | 0        | 0        | 0        | 2.0         | 3.0         | 67%        |
| 4.4.2 <a href="#">Configure IPv4 iptables</a>                            | 2         | 1         | 0        | 0        | 1        | 0        | 2.0         | 3.0         | 67%        |
| 4.4.3 <a href="#">Configure IPv6 ip6tables</a>                           | 2         | 1         | 0        | 0        | 1        | 0        | 2.0         | 3.0         | 67%        |
| <b>5 Access Control</b>                                                  | <b>59</b> | <b>7</b>  | <b>0</b> | <b>0</b> | <b>1</b> | <b>0</b> | <b>59.0</b> | <b>66.0</b> | <b>89%</b> |
| 5.1 <a href="#">Configure SSH Server</a>                                 | 22        | 0         | 0        | 0        | 0        | 0        | 22.0        | 22.0        | 100%       |
| 5.2 <a href="#">Configure privilege escalation</a>                       | 6         | 0         | 0        | 0        | 0        | 0        | 6.0         | 6.0         | 100%       |
| 5.3 <a href="#">Pluggable Authentication Modules</a>                     | 17        | 6         | 0        | 0        | 1        | 0        | 17.0        | 23.0        | 74%        |
| 5.3.1 <a href="#">Configure PAM software packages</a>                    | 3         | 0         | 0        | 0        | 0        | 0        | 3.0         | 3.0         | 100%       |
| 5.3.2 <a href="#">Configure pam-auth-update profiles</a>                 | 3         | 1         | 0        | 0        | 0        | 0        | 3.0         | 4.0         | 75%        |
| 5.3.3 <a href="#">Configure PAM Arguments</a>                            | 11        | 5         | 0        | 0        | 1        | 0        | 11.0        | 16.0        | 69%        |
| 5.3.3.1 <a href="#">Configure pam_faillock module</a>                    | 0         | 2         | 0        | 0        | 0        | 0        | 0.0         | 2.0         | 0%         |
| 5.3.3.2 <a href="#">Configure pam_pwquality module</a>                   | 6         | 1         | 0        | 0        | 1        | 0        | 6.0         | 7.0         | 86%        |
| 5.3.3.3 <a href="#">Configure pam_pwhistory module</a>                   | 2         | 1         | 0        | 0        | 0        | 0        | 2.0         | 3.0         | 67%        |
| 5.3.3.4 <a href="#">Configure pam_unix module</a>                        | 3         | 1         | 0        | 0        | 0        | 0        | 3.0         | 4.0         | 75%        |
| 5.4 <a href="#">User Accounts and Environment</a>                        | 14        | 1         | 0        | 0        | 0        | 0        | 14.0        | 15.0        | 93%        |
| 5.4.1 <a href="#">Configure shadow password suite parameters</a>         | 4         | 1         | 0        | 0        | 0        | 0        | 4.0         | 5.0         | 80%        |
| 5.4.2 <a href="#">Configure root and system accounts and environment</a> | 8         | 0         | 0        | 0        | 0        | 0        | 8.0         | 8.0         | 100%       |

| Description                                                    | Tests |      |       |       |      |      | Scoring |       |         |
|----------------------------------------------------------------|-------|------|-------|-------|------|------|---------|-------|---------|
|                                                                | Pass  | Fail | Error | Unkn. | Man. | Exc. | Score   | Max   | Percent |
| 5.4.3 <a href="#">Configure user default environment</a>       | 2     | 0    | 0     | 0     | 0    | 0    | 2.0     | 2.0   | 100%    |
| 6 <a href="#">Logging and Auditing</a>                         | 14    | 2    | 0     | 0     | 6    | 0    | 14.0    | 16.0  | 88%     |
| 6.1 <a href="#">Configure Integrity Checking</a>               | 1     | 2    | 0     | 0     | 0    | 0    | 1.0     | 3.0   | 33%     |
| 6.2 <a href="#">System Logging</a>                             | 13    | 0    | 0     | 0     | 6    | 0    | 13.0    | 13.0  | 100%    |
| 6.2.1 <a href="#">Configure systemd-journald service</a>       | 1     | 0    | 0     | 0     | 2    | 0    | 1.0     | 1.0   | 100%    |
| 6.2.2 <a href="#">Configure journald</a>                       | 6     | 0    | 0     | 0     | 1    | 0    | 6.0     | 6.0   | 100%    |
| 6.2.2.1 <a href="#">Configure systemd-journal-remote</a>       | 3     | 0    | 0     | 0     | 1    | 0    | 3.0     | 3.0   | 100%    |
| 6.2.3 <a href="#">Configure rsyslog</a>                        | 5     | 0    | 0     | 0     | 3    | 0    | 5.0     | 5.0   | 100%    |
| 6.2.4 <a href="#">Configure Logfiles</a>                       | 1     | 0    | 0     | 0     | 0    | 0    | 1.0     | 1.0   | 100%    |
| 6.3 <a href="#">System Auditing</a>                            | 0     | 0    | 0     | 0     | 0    | 0    | 0.0     | 0.0   | 0%      |
| 6.3.1 <a href="#">Configure auditd Service</a>                 | 0     | 0    | 0     | 0     | 0    | 0    | 0.0     | 0.0   | 0%      |
| 6.3.2 <a href="#">Configure Data Retention</a>                 | 0     | 0    | 0     | 0     | 0    | 0    | 0.0     | 0.0   | 0%      |
| 6.3.3 <a href="#">Configure auditd Rules</a>                   | 0     | 0    | 0     | 0     | 0    | 0    | 0.0     | 0.0   | 0%      |
| 6.3.4 <a href="#">Configure auditd File Access</a>             | 0     | 0    | 0     | 0     | 0    | 0    | 0.0     | 0.0   | 0%      |
| 7 <a href="#">System Maintenance</a>                           | 21    | 1    | 0     | 0     | 1    | 0    | 21.0    | 22.0  | 95%     |
| 7.1 <a href="#">Configure system file and directory access</a> | 12    | 0    | 0     | 0     | 1    | 0    | 12.0    | 12.0  | 100%    |
| 7.2 <a href="#">Local User and Group Settings</a>              | 9     | 1    | 0     | 0     | 0    | 0    | 9.0     | 10.0  | 90%     |
| Total                                                          | 194   | 34   | 0     | 0     | 19   | 0    | 194.0   | 228.0 | 85%     |

**Note:** Actual scores are subject to rounding errors. The sum of these values may not result in the exact overall score.

The 'Exc' column only applies to Exceptions that are generated using CIS-CAT Pro Dashboard and is not utilized by CIS-CAT Pro Assessor.

## Profiles

This benchmark contains 4 profiles. The **Level 1 - Workstation** profile was used for this assessment.

| Title                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Level 1 - Server      | <p>Items in this profile intend to:</p> <ul style="list-style-type: none"> <li>be practical and prudent;</li> <li>provide a clear security benefit; and</li> <li>not inhibit the utility of the technology beyond acceptable means.</li> </ul> <p>This profile is intended for servers.</p> <p><a href="#">Show Profile XML</a></p>                                                                                                                                                       |
| Level 2 - Server      | <p>This profile extends the "Level 1 - Server" profile. Items in this profile exhibit one or more of the following characteristics:</p> <ul style="list-style-type: none"> <li>are intended for environments or use cases where security is paramount.</li> <li>acts as defense in depth measure.</li> <li>may negatively inhibit the utility or performance of the technology.</li> </ul> <p>This profile is intended for servers.</p> <p><a href="#">Show Profile XML</a></p>           |
| Level 1 - Workstation | <p>Items in this profile intend to:</p> <ul style="list-style-type: none"> <li>be practical and prudent;</li> <li>provide a clear security benefit; and</li> <li>not inhibit the utility of the technology beyond acceptable means.</li> </ul> <p>This profile is intended for workstations.</p> <p><a href="#">Show Profile XML</a></p>                                                                                                                                                  |
| Level 2 - Workstation | <p>This profile extends the "Level 1 - Workstation" profile. Items in this profile exhibit one or more of the following characteristics:</p> <ul style="list-style-type: none"> <li>are intended for environments or use cases where security is paramount.</li> <li>acts as defense in depth measure.</li> <li>may negatively inhibit the utility or performance of the technology.</li> </ul> <p>This profile is intended for workstations.</p> <p><a href="#">Show Profile XML</a></p> |